

Geant4-TCAD Simulation of 4H-SiC PIN Detectors for C-14 Beta Sources

Abstract

This work evaluates whether monoenergetic electrons can replace the real C-14 beta spectrum in 4H-SiC PIN detector simulations and uses the result for detector design. A coupled Geant4-Sentaurus workflow converts energy deposition into TCAD carrier generation maps, from which transient current, CCE, peak current, and t90 are extracted. The results show that monoenergetic sources may approximate some charge metrics but cannot reproduce the real C-14 transient i-t response, including waveform shape and peak current. The detector is therefore optimized directly under the real C-14 spectrum using a full-depletion thickness scan from 10 to 130 μm . With a CCE-t90 selection rule, the recommended C-14 design is a 60 μm i region biased at 19 V, giving CCE of 97.332 percent and t90 of 6.559 ns. Extending the same rule to 20, 49, 100, and 156.5 keV sources shows that the optimal detector thickness is source dependent.

Keywords: 4H-SiC, PIN detector, C-14 beta spectrum, Geant4, Sentaurus TCAD, monoenergetic approximation, charge collection efficiency

1. Introduction

1.1 Background

Wide-bandgap semiconductor radiation detectors are attractive for operation in high-temperature, high-radiation, and low-leakage environments. Compared with conventional silicon detectors, 4H-SiC has a bandgap of about 3.26 eV, which suppresses room-temperature dark current. Its high critical electric field enables operation under large reverse bias, while its displacement threshold energy and chemical stability support long-term operation in nuclear radiation environments. These properties make 4H-SiC PIN diodes a suitable platform for modeling C-14 beta response and for optimizing detector structures.

C-14 undergoes beta decay according to

$$^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \bar{\nu}_e$$

Because the antineutrino carries a variable fraction of the decay energy, the emitted electron does not have a single energy. The C-14 beta spectrum extends from near zero to an endpoint energy of 156.5 keV, with a mean energy of about 49 keV. In a 4H-SiC PIN detector, this continuous spectrum produces a spatial generation profile that combines near-surface deposition from low-energy electrons and a deeper tail from high-energy electrons. The real beta spectrum therefore affects not only the total generated charge but also the carrier drift path, transient current waveform, peak current, and collection time.

1.2 Scientific Problem

In Geant4-TCAD simulations and detector-response calculations, the real beta spectrum is often replaced by a monoenergetic electron at the mean energy or at the endpoint energy to reduce simulation complexity. This simplification may be acceptable for some integral quantities, such as total deposited energy or total collected charge. However, the transient current response depends

on the depth distribution of carrier generation, local electric field, drift distance, and carrier arrival time distribution.

Two questions follow. First, under what conditions can a monoenergetic source replace the real C-14 spectrum in detector simulations. Second, if the real spectrum cannot be reliably replaced for transient response, should the detector i-region thickness and full-depletion bias be designed directly for the real spectrum rather than extrapolated from a representative monoenergetic case.

1.3 Contributions

This work makes four contributions.

1. A coupled Geant4-Sentaurus workflow is established to convert energy deposition from the real C-14 beta spectrum and selected monoenergetic electron sources into TCAD carrier generation maps. CCE, peak current, t90, and FWHM are extracted under a unified geometry and post-processing procedure.
2. The boundary of monoenergetic approximation is clarified. Some monoenergetic sources can give CCE values close to the real C-14 result, but they do not reliably replace the real spectrum for transient i-t waveforms, peak current, or charge collection time.
3. A full-depletion 4H-SiC PIN design scan is performed directly under the real C-14 spectrum. The scan uses CCE and t90 as joint criteria and identifies the recommended i-region thickness for the real beta source.
4. The same CCE-t90 design rule is extended to selected monoenergetic sources. The resulting recommended detector parameters show that the optimal thickness depends strongly on the source spectrum and deposition depth distribution.

2. Device Structure and Coupled Geant4-Sentaurus Method

2.1 4H-SiC PIN Detector Structure

The studied detector is a vertically irradiated 4H-SiC PIN diode. It consists of a top p+ contact layer, a lightly doped i region, and a bottom n+ contact layer. Electrons enter from the p+ side. The baseline structure uses a 0.2 μm p+ layer, a 120 μm i region, and a 0.5 μm n+ layer. The net donor concentration in the i region is $5.6 \times 10^{12} \text{ cm}^{-3}$, and the p+ and n+ contact regions are heavily doped at $1 \times 10^{19} \text{ cm}^{-3}$.

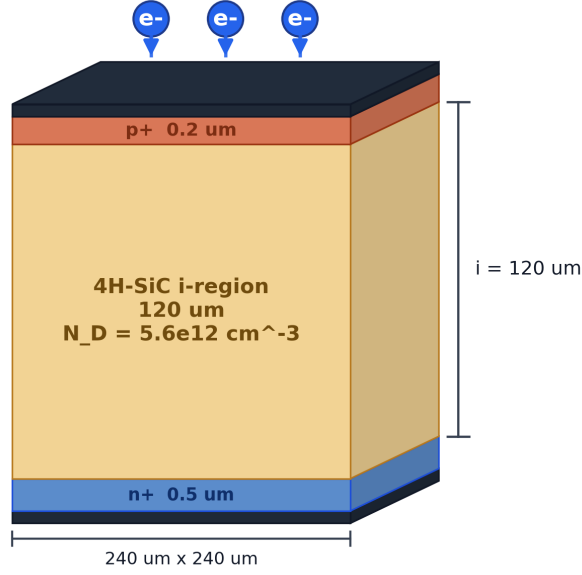


Fig. 1. Schematic structure of the 4H-SiC PIN detector. C-14 beta electrons are incident from the p+ side, and the device operates under reverse bias.

Table 1. Device and material parameters.

Parameter	Value
p+ layer thickness	0.2 um
Baseline i-region thickness	120 um
n+ layer thickness	0.5 um
i-region net donor concentration	$5.6 \times 10^{12} \text{ cm}^{-3}$
p+ and n+ doping concentration	$1 \times 10^{19} \text{ cm}^{-3}$
Depletion voltage for 120 um i region	about 75 V

For different i-region thicknesses, the one-dimensional depletion approximation gives

$$V_{\text{dep}}(W_i) = \frac{qN_D W_i^2}{2\epsilon_{\text{SiC}}}$$

Using the 120 um device as a reference, this becomes

$$V_{\text{dep}}(W_i) = 75 \text{ V} \left(\frac{W_i}{120 \text{ } \mu\text{m}} \right)^2$$

Each thickness in the subsequent full-depletion scan is therefore evaluated at its corresponding depletion voltage rather than at a fixed external bias. This treatment ensures that different thicknesses are compared under comparable depletion conditions.

2.2 Coupling from Geant4 to Sentaurus

Geant4 is used to calculate the energy deposition of each source term inside the 4H-SiC PIN structure. The deposition profile is then converted into a spatial electron-hole pair generation map that can be imported into Sentaurus TCAD. The workflow is shown in Fig. 2.



Fig. 2. Coupled Geant4-Sentaurus simulation workflow. The source term is either the real C-14 beta spectrum or a monoenergetic electron source. Geant4 provides the deposited-energy distribution, which is converted into a carrier generation map and imported into Sentaurus for transient current simulation.

For each source term, the Geant4 energy deposition is normalized by the number of incident particles and converted into electron-hole pairs using an average pair creation energy of $E_{eh} = 7.8$ eV.

Sentaurus solves the transient cathode current $i(t)$ using the same detector geometry, mesh strategy, and transient solver settings for all source terms. The collected charge is calculated as

$$Q_{\text{collect}} = \int i(t) dt$$

The charge collection efficiency used in this work is

$$\text{CCE} = \frac{Q_{\text{collect}}}{Q_{\text{gen,ref}}}$$

Here $Q_{\text{gen,ref}}$ is the total generated charge in the reference generation map for the corresponding source term. This definition is used throughout the manuscript. It captures both incomplete charge collection and loss caused by truncating the deposition profile in a finite-thickness detector. It is therefore different from an internal collection efficiency normalized only by the charge generated inside the current detector thickness.

The charge collection time t_{90} is defined by

$$\int_0^{t_{90}} i(t) dt = 0.9Q_{\text{collect}}$$

This definition measures the time required for the cumulative collected charge to reach 90 percent of the final collected charge. It is more appropriate for readout-window analysis than a waveform-amplitude decay time.

3. Boundary of Monoenergetic Approximation for the Real C-14 Spectrum

3.1 C-14 Beta Spectrum

The C-14 beta decay produces a continuous electron spectrum. Its probability density is determined by the electron momentum, total energy, remaining decay energy, and Coulomb correction. Fig. 3 shows the C-14 beta spectrum used in this work and the corresponding Geant4 sampling result.

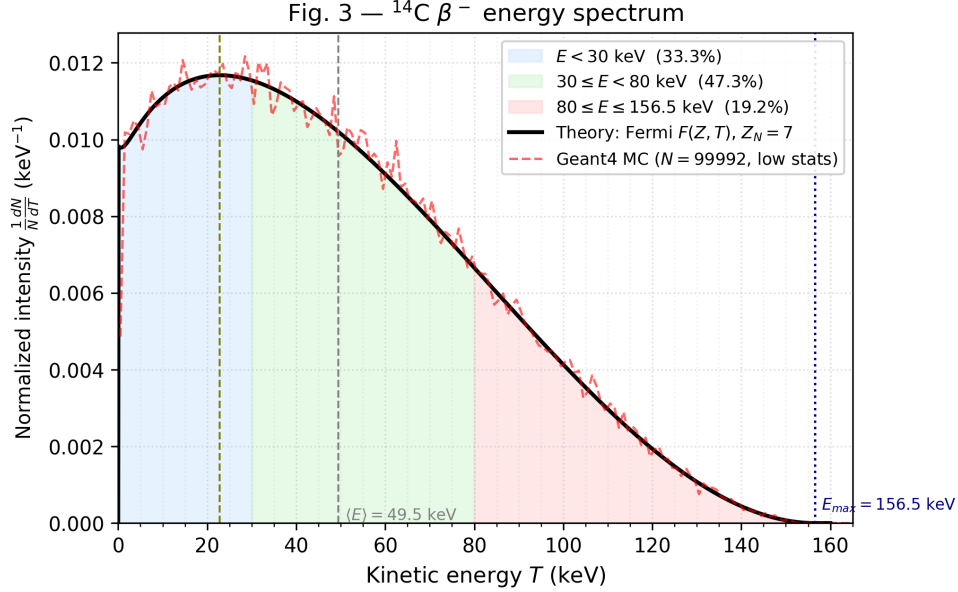


Fig. 3. C-14 beta spectrum. The spectrum is continuous, with a mean energy of about 49 keV and an endpoint energy of 156.5 keV.

The continuous spectrum implies that different electron energies contribute to different deposition depths in 4H-SiC. Low-energy electrons deposit their energy close to the incident surface, while high-energy electrons create a deeper deposition tail. A single monoenergetic electron can reproduce only one part of this spatial behavior.

3.2 Depth Distribution of Energy Deposition

Fig. 4 compares the depth-dependent energy deposition profiles of the real C-14 spectrum and monoenergetic electrons at 20, 49, 100, and 156.5 keV.

Fig. 4 — Energy deposition depth profile in 4H-SiC PIN

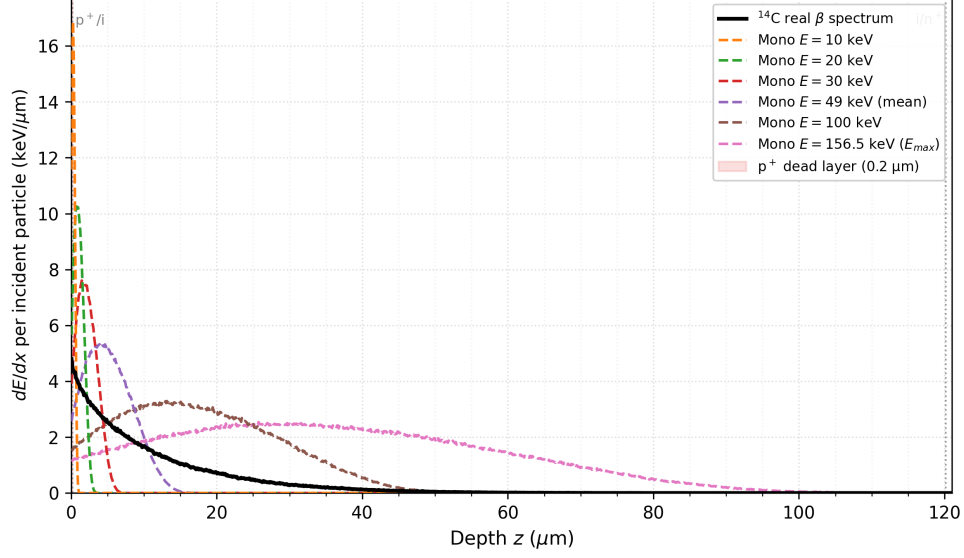


Fig. 4. Depth-dependent energy deposition for different source terms in 4H-SiC. The real C-14 spectrum contains both shallow deposition and a deeper high-energy tail, whereas each monoenergetic source represents only a limited portion of the spectrum.

The 20 keV source deposits energy mainly near the entrance side and is sensitive to the p+ dead layer and shallow i-region. The 49 keV source is close to the mean beta energy, but its deposition remains concentrated in a limited depth range. The 100 and 156.5 keV sources penetrate more deeply and enhance the deposition tail. The real C-14 profile is a weighted superposition over the full spectrum and is therefore not equivalent to either the mean-energy or endpoint-energy source.

3.3 Transient i-t Response

To evaluate the approximation boundary, TCAD transient simulations are performed for 20 keV, 49 keV, the real C-14 spectrum, 100 keV, and 156.5 keV under the same 4H-SiC PIN structure, reverse bias, and post-processing method. The resulting cathode current waveforms are shown in Fig. 5.

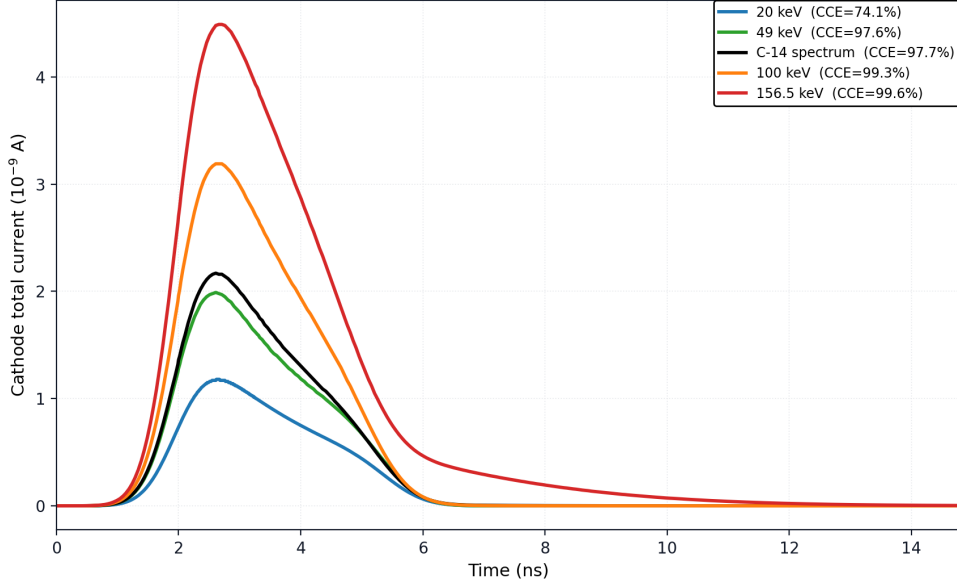


Fig. 5. Transient cathode current responses for 20 keV, 49 keV, the real C-14 spectrum, 100 keV, and 156.5 keV inputs. Similar CCE values do not imply equivalent peak current, waveform shape, or charge collection time.

Table 2. Comparison of CCE, peak current, and t90 between monoenergetic sources and the real C-14 spectrum.

Input source	CCE (%)	I _{peak} (nA)	t ₉₀ (ns)
20 keV	74.08	1.180	4.875
49 keV	97.62	1.990	4.809
C-14 spectrum	97.66	2.170	4.763
100 keV	99.28	3.190	4.671
156.5 keV	99.56	4.490	5.532

If only CCE is considered, the 49 keV monoenergetic electron appears close to the real C-14 spectrum. Their CCE values differ by only about 0.04 percentage points. This observation indicates that monoenergetic approximation should not be rejected solely on the basis of integral charge metrics.

The transient response gives a different conclusion. The peak current of the real C-14 spectrum is about 2.170 nA, whereas the 49 keV source gives 1.990 nA, the 100 keV source gives 3.190 nA, and the 156.5 keV source gives 4.490 nA. The endpoint-energy source also increases t₉₀ to 5.532 ns. Thus, a mean-energy source underestimates the peak current, while high-energy monoenergetic sources overestimate it and change the waveform.

CCE is an integrated quantity and can hide differences in carrier generation depth and arrival-time distribution. For front-end readout design, integration-window selection, and pulse-shape analysis, the real C-14 spectrum cannot be reliably replaced by a monoenergetic electron source. The first main conclusion is therefore that monoenergetic approximation is useful for limited integral estimates but not for transient i-t response analysis.

4. Full-Depletion Design for the Real C-14 Spectrum

The previous section shows that the real C-14 transient response cannot be represented reliably by a single monoenergetic source. Detector optimization for C-14 detection should therefore use the real beta spectrum directly rather than a mean-energy or endpoint-energy surrogate.

4.1 Full-Depletion Bias Normalization

The i-region thickness is scanned from 10 to 130 μm in 10 μm steps. The corresponding depletion voltages are 1, 3, 5, 9, 14, 19, 26, 33, 42, 52, 63, 75, and 88 V.

These values follow from

$$V_{\text{dep}}(W_i) = 75 \text{ V} \left(\frac{W_i}{120 \mu\text{m}} \right)^2$$

Each thickness is therefore evaluated near its full-depletion condition. This prevents the comparison from being biased by different degrees of depletion at different detector thicknesses.

4.2 Thickness-Scan Results under the C-14 Spectrum

Fig. 6a and Fig. 6b show the full-depletion thickness scan for all source terms. The C-14 curve is first used to determine the real-spectrum detector design.

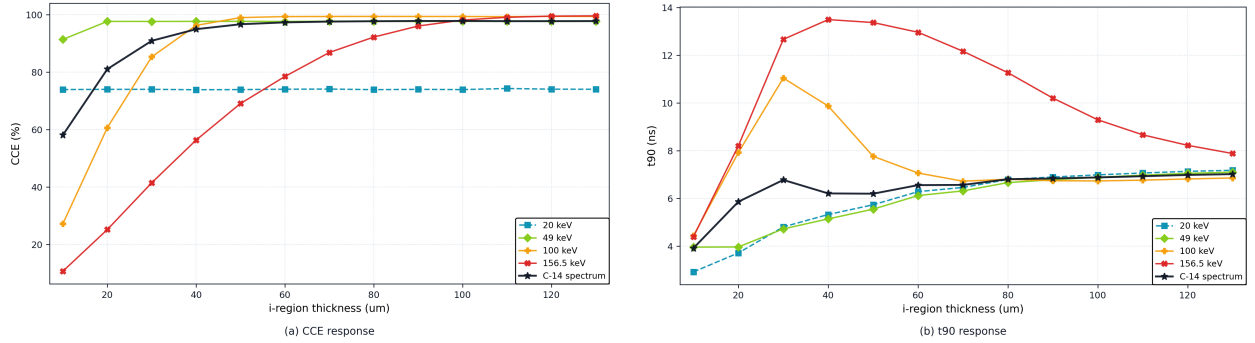


Fig. 6. Full-depletion thickness scan for selected monoenergetic sources and the real C-14 spectrum. Fig. 6a gives CCE as a function of i-region thickness. Fig. 6b gives t90 as a function of i-region thickness.

Table 3. Full-depletion metrics for the real C-14 spectrum at different i-region thicknesses.

W_i (μm)	V_{dep} (V)	CCE (%)	t90 (ns)	I_{peak} (nA)
10	1	58.156	3.918	2.127
20	3	81.075	5.864	2.145
30	5	90.927	6.777	1.781
40	9	94.965	6.211	1.753
50	14	96.685	6.200	1.740
60	19	97.332	6.559	1.688
70	26	97.610	6.569	1.701
80	33	97.738	6.809	1.681
90	42	97.811	6.839	1.687

Wi (um)	Vdep (V)	CCE (%)	t90 (ns)	Ipeak (nA)
100	52	97.828	6.882	1.690
110	63	97.757	6.933	1.688
120	75	97.759	6.986	1.686
130	88	97.785	7.021	1.686

For the real C-14 spectrum, CCE increases rapidly with i-region thickness. The 10 um detector collects only 58.156 percent of the reference generated charge, and the 20 and 30 um detectors still suffer from significant truncation of the deposition profile. At 50 um, the CCE reaches 96.685 percent. At 60 um, it further increases to 97.332 percent. From 70 to 130 um, CCE enters a high-efficiency plateau, with a maximum value of about 97.828 percent.

The t90 trend does not monotonically favor thicker detectors. A thicker i region covers a deeper deposition tail but also increases the carrier drift distance. For the real C-14 spectrum, t90 is about 6.57 to 7.02 ns for thicknesses from 70 to 130 um, while the 60 um detector already lies near the high-CCE plateau and keeps a comparable response time.

4.3 Recommended C-14 i-Region Parameter

The design rule used here is as follows. First, the maximum CCE under the given source term is identified. Second, all thicknesses with CCE within 0.5 percentage points of that maximum are retained. Third, the candidate with the shortest t90 is selected.

For the C-14 spectrum, the maximum CCE is about 97.828 percent. The 0.5 percentage-point window gives a threshold of 97.328 percent. The 60 um detector reaches 97.332 percent, just entering this high-CCE candidate region, and it has a shorter t90 than the thicker candidates. The recommended full-depletion structure is therefore $W_i = 60 \mu\text{m}$ and $V_{\text{dep}} = 19 \text{ V}$. Its main performance metrics are CCE of 97.332 percent, t90 of 6.559 ns, and $I_{\text{peak}} = 1.688 \text{ nA}$.

This point is not the absolute maximum-CCE design. It is a practical compromise that preserves near-plateau charge collection while avoiding unnecessary increase in collection time.

5. Extension of the Design Rule to Monoenergetic Sources

5.1 Multi-Source Full-Depletion Thickness Scan

The same full-depletion thickness scan and CCE-t90 selection rule are applied to four monoenergetic sources: 20, 49, 100, and 156.5 keV.

The curves in Fig. 6 show strong source dependence. As the incident electron energy increases, the deposition tail becomes deeper, and a thicker i region is generally required to reach high CCE. The real C-14 spectrum, containing both shallow and deep contributions, gives an intermediate recommended thickness.

The most important comparison is between the real C-14 spectrum and the 49 keV monoenergetic source. Although 49 keV is close to the mean C-14 beta energy, its recommended thickness is only 20 um, whereas the real C-14 spectrum requires 60 um. Thus, the mean-energy monoenergetic source does not represent the thickness requirement imposed by the high-energy tail of the real spectrum.

5.2 Recommended Design Points

Fig. 7 summarizes the recommended i-region thickness, depletion voltage, CCE, and t90 for each source term.

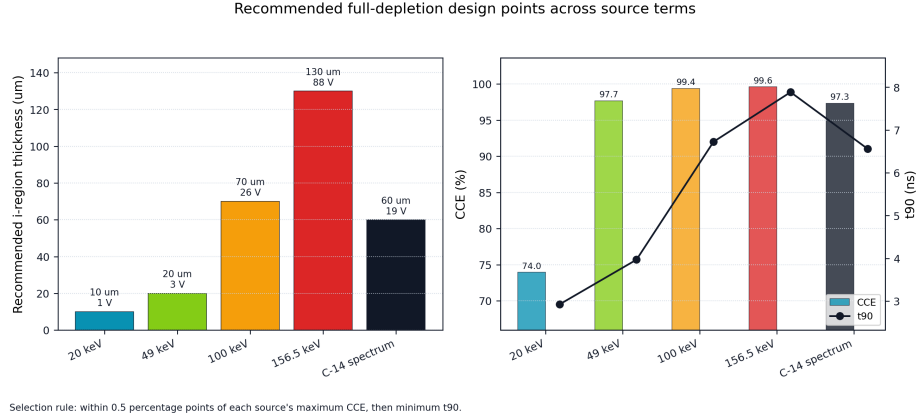


Fig. 7. Recommended full-depletion design points for different source terms. For each source, the selected point lies within 0.5 percentage points of the maximum CCE and has the shortest t90 among the candidates.

Table 4. Recommended detector parameters for different source terms.

Source	Wi (um)	Vdep (V)	CCE (%)	t90 (ns)	Ipeak (nA)
20 keV	10	1	73.952	2.922	2.609
49 keV	20	3	97.677	3.968	2.755
100 keV	70	26	99.384	6.725	2.318
156.5 keV	130	88	99.600	7.885	3.306
C-14 spectrum	60	19	97.332	6.559	1.688

The recommended thickness increases with source energy in general. The 20 keV source selects a 10 um i region, while the 49 keV source selects 20 um. The 100 keV source selects 70 um, and the 156.5 keV endpoint source selects 130 um. These values reflect the increasing penetration depth of higher-energy electrons.

The real C-14 spectrum selects 60 um, which is much thicker than the 49 keV mean-energy design but thinner than the endpoint-energy design. This behavior is consistent with the physical spectrum: most electrons contribute shallow deposition, while the high-energy tail still requires a finite active thickness. The result confirms that real-spectrum-driven design cannot be replaced by direct extrapolation from a single monoenergetic energy.

6. Discussion

The results demonstrate that the validity of monoenergetic approximation depends on the response metric. CCE is obtained by integrating the current waveform and therefore does not preserve the temporal arrival order of charge. Two source terms may produce similar CCE while generating different peak currents, pulse widths, and t90 values. For applications that only require final count or total collected charge, a monoenergetic approximation may have engineering value. For front-end

electronics design, timing-window selection, and pulse-shape analysis, the real source spectrum is necessary.

The C-14 optimization also shows why a thickness design based only on t_{90} is not sufficient. Very thin detectors can be fast, but they truncate the real C-14 deposition profile and reduce CCE. Conversely, very thick detectors approach the CCE plateau but increase drift distance and collection time. The adopted rule, choosing the shortest t_{90} within a near-maximum CCE window, provides a practical tradeoff between charge completeness and transient speed. The 0.5 percentage-point window used here is an engineering choice for demonstrating the method and can be adjusted according to readout noise, timing resolution, minimum signal amplitude, or system-level detection requirements.

The difference between the real C-14 design and the 49 keV design is central to the paper. The mean energy captures only one spectral statistic and does not encode the spatial deposition distribution. The optimal detector thickness is governed by the spectrum-dependent generation depth profile, not by the mean energy alone.

Low-energy detection is left as a future research direction rather than the main line of this paper. Future work can extend the same Geant4-Sentaurus and CCE- t_{90} workflow to thinner dead layers, entrance-window materials, readout capacitance, leakage current, electronic noise, and experimental process constraints.

7. Conclusion

This work establishes a coupled Geant4-Sentaurus simulation workflow for C-14 beta response in 4H-SiC PIN detectors and uses it to evaluate the boundary of monoenergetic approximation and to guide full-depletion detector design.

First, monoenergetic approximation has a clear boundary. The 49 keV source gives a CCE close to that of the real C-14 spectrum, but it does not reliably reproduce the transient i - t waveform, peak current, or t_{90} . Monoenergetic sources should therefore not be used as direct substitutes for real-spectrum transient-response analysis.

Second, a real-C-14-driven full-depletion thickness scan identifies a practical detector design. Under the adopted selection rule, the recommended parameters are $W_i = 60 \mu\text{m}$ and $V_{\text{dep}} = 19 \text{ V}$, with CCE of 97.332 percent and t_{90} of 6.559 ns.

Third, the same design rule is extended to selected monoenergetic sources. The recommended i -region thickness varies strongly with incident energy and deposition depth, ranging from 10 μm for 20 keV to 130 μm for 156.5 keV. The real C-14 recommended thickness differs from the 49 keV mean-energy design, further confirming the need for real-spectrum-driven detector optimization.

Overall, the proposed workflow provides a complete route from realistic beta-spectrum transport to TCAD transient response and full-depletion parameter selection. It can support the structural optimization of 4H-SiC PIN detectors for C-14 beta-source applications and provide input for subsequent readout-circuit matching.

References

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